phenomenon of diffraction in this chapter and will show that the spreading out decreases with decrease in wavelength. Indeed, since the light wavelengths are very small ($\lambda \sim 5 \times 10^{-5}$ cm), the effects due to diffraction are not readily observed.



Fig. 18.1 If a plane wave is incident on an aperture then according to geometrical optics a sharp shadow will be cast in the region AB of the screen.

We should point out that there is not much of a difference between the phenomena of interference and diffraction, indeed, interference corresponds to the situation when we consider the superposition of waves coming out from a number of point sources and diffraction corresponds to the situation when we consider waves coming out from an area source like a circular or rectangular aperture or even a large number of rectangular apertures (like the diffraction grating).

The diffraction phenomena are usually divided into two categories: (i) Fresnel diffraction and (ii) Fraunhofer diffraction.

In the Fresnel class of diffraction, the source of light and the screen are, in general, at a finite distance from the diffracting aperture [see Fig. 18.2(a)]. In the Fraunhofer class of diffraction, the source and the screen are at infinite distances from the aperture; this is easily achieved by placing the source on the focal plane of a convex lens and placing the screen on the focal plane of another convex lens [see Fig. 18.2(b)]. The two lenses effectively moved the source and the screen to infinity because the first lens makes the light beam parallel and the second lens effectively makes the screen receive a parallel beam of light. It turns out that it is much easier to calculate the intensity distribution of a Fraunhofer diffraction pattern which we plan to do in this chapter. Further, the Fraunhofer diffraction pattern is not difficult to observe; all that one needs is an ordinary laboratory spectrometer; the collimator renders a parallel beam of light and the telescope receives parallel beams of light on its focal plane. The diffracting aperture is placed on the prism table. In Chapter 20, we will study the Fresnel class of diffraction and will discuss the transition from the Fresnel region to the Fraunhofer region.

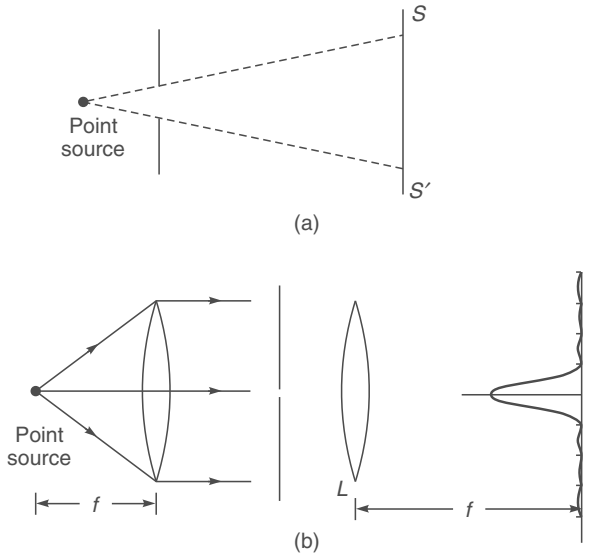


Fig. 18.2 (a) When either the source or the screen (or both) are at finite distances from the aperture, the diffraction pattern corresponds to the Fresnel class. (b) In the Fraunhofer class both the source and the screen are at infinity.

18.2 SINGLE-SLIT DIFFRACTION PATTERN

LO 2

We will first study the Fraunhofer diffraction pattern produced by an infinitely long slit of width b. A plane wave is assumed to fall normally on the slit and we wish to calculate the intensity distribution on the focal plane of the lens L [see Fig. 18.3(a)]. We assume that the slit consists of a large number of equally spaced point sources and that each point on the slit is a source of Huygens' secondary wavelets which interfere with the wavelets emanating from other points. Let the point sources be at $A_1, A_2, A_3, \ldots$ and let the distance between two consecutive points be Δ [see Fig. 18.3(b)]. Thus, if the number of point sources be n, then

$$b = (n-1)\Delta \qquad \textbf{(18.1)}$$

We will now calculate the resultant field produced by these n sources at the point P, P being an arbitrary point (on the focal plane of the lens) receiving parallel rays making an angle θ with the normal to the slit [see Fig. 18.3(b)]. Since the slit actually consists of a continuous distribution of sources, we will, in the final expression, let n go to infinity and Δ go to zero such that $n\Delta$ tends to b.

Now, at the point P, the amplitudes of the disturbances reaching from $A_1, A_2, \ldots$ will be very nearly the same because the point P is at a distance which is very large in comparison to b [see Fig. 18.3(b)]. However, because of even slightly different path lengths to the point P, the field produced by A_1 will differ in phase from the field produced by A_2.

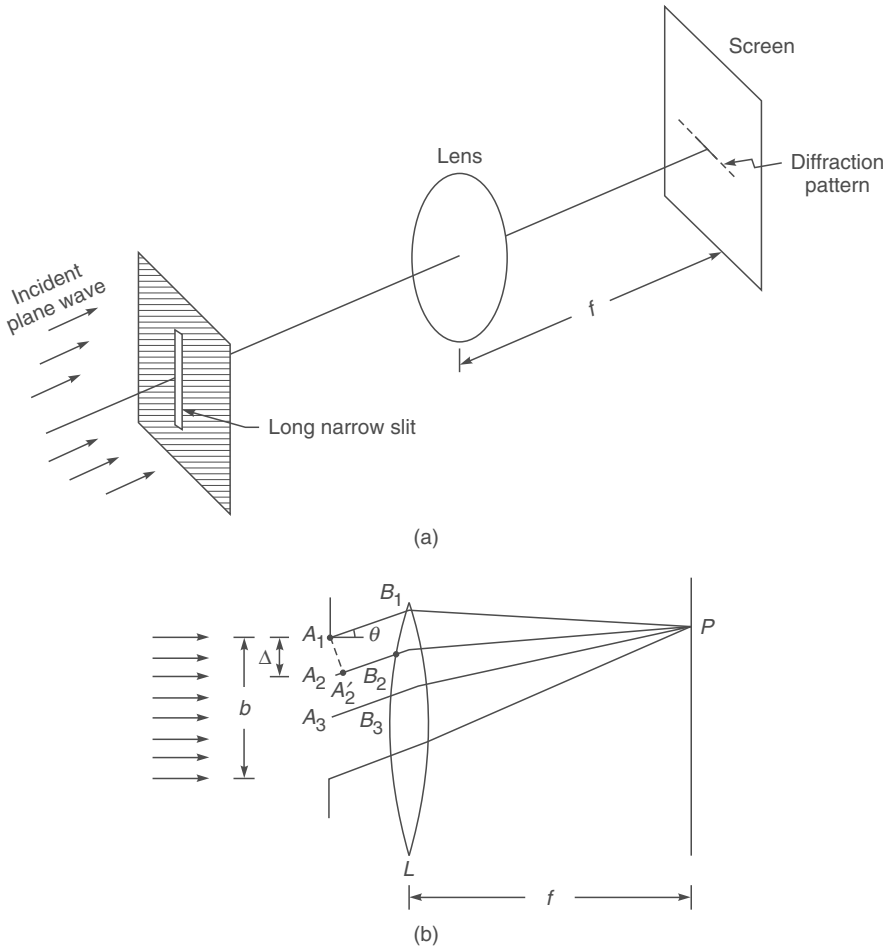


Fig. 18.3 (a) Diffraction of a plane wave incident normally on a long narrow slit of width b. Notice that the spreading occurs along the width of the slit. (b) In order to calculate the diffraction pattern, the slit is assumed to consist of a large number of equally spaced points.

For an incident plane wave, the points A_1, A_2,... are in phase and, therefore, the additional path traversed by the disturbance emanating from the point A_2 will be A_2A_2' where A_2' is the foot of the perpendicular drawn from A_1 on A_2B_2. This follows from the fact that the optical paths A_1B_1P and $A_2'B_2P$ are the same. If the diffracted rays make an angle θ with the normal to the slit then the path difference would be

$$A_2A_2' = \Delta \sin \theta$$

The corresponding phase difference, ϕ, would be given by

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta \qquad \textbf{(18.2)}$$

Thus, if the field at the point P due to the disturbance emanating from the point A_1 is $a \cos \omega t$ then the field due to the disturbance emanating from A_2 would be $a \cos (\omega t - \phi)$. Now the difference in the phases of the disturbance reaching from the points A_2 and A_3 will also be ϕ and thus the resultant field at the point P would be given by

$$E = a[\cos \omega t + \cos (\omega t - \phi) + \ldots + \cos (\omega t - (n-1)\phi)] \qquad \textbf{(18.3)}$$

where

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta$$

Now, we had shown in Sec. 11.7 that

$$\cos \omega t + \cos (\omega t - \phi) + \ldots + \cos [\omega t - (n-1)\phi]$$
$$= \frac{\sin n\phi/2}{\sin \phi/2} \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right] \qquad \textbf{(18.4)}$$

Thus,

$$E = E_0 \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right] \qquad \textbf{(18.5)}$$

where the amplitude E_θ of the resultant field would be given by*

$$E_\theta = a\frac{\sin(n\phi/2)}{\sin\phi/2} \tag{18.6}$$

In the limit of $n \to \infty$ and $\Delta \to 0$ in such a way that $n\Delta \to b$, we have

$$\frac{n\phi}{2} = \frac{\pi}{\lambda} n\Delta \sin\theta \to \frac{\pi}{\lambda} b \sin\theta$$

Further,

$$\phi = \frac{2\pi}{\lambda}\Delta \sin\theta = \frac{2\pi}{\lambda}\frac{b\sin\theta}{n}$$

would tend to zero and we may, therefore, write

$$E_\theta \approx \frac{a\sin\left(\frac{n\phi}{2}\right)}{\frac{\phi}{2}} = na\,\frac{\sin\frac{\pi b\sin\theta}{\lambda}}{\frac{\pi b\sin\theta}{\lambda}}$$

$$= A\,\frac{\sin\beta}{\beta} \tag{18.7}$$

where**

$$A = na$$

and

$$\beta = \frac{\pi b\sin\theta}{\lambda} \tag{18.8}$$

Thus,

$$E = A\,\frac{\sin\beta}{\beta}\cos(\omega t - \beta) \tag{18.9}$$

The corresponding intensity distribution is given by

$$I = I_0\,\frac{\sin^2\beta}{\beta^2} \tag{18.10}$$

where I_0 represents the intensity at $\theta = 0$

18.2.1 Positions of Maxima and Minima

The variation of the intensity with β is shown in Fig. 18.4(a). It is obvious from Eq. (18.10) that the intensity is zero when

$$\beta = m\pi,\ m \neq 0 \tag{18.11}$$

[When $\beta = 0$, $\frac{\sin\beta}{\beta} = 1$ and $I = I_0$ which corresponds to the maximum of the intensity.] Substituting the value of β one obtains

$$b\sin\theta = m\lambda;\ m = \pm 1, \pm 2, \pm 3, \ldots \text{ (minima)} \tag{18.12}$$

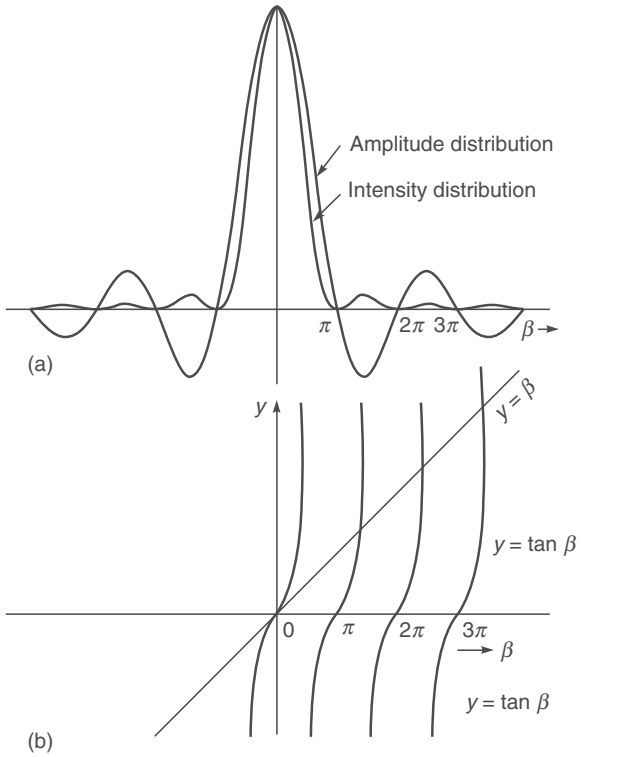


Fig. 18.4 (a) The intensity distribution corresponding to the single slit Fraunhofer diffraction pattern. (b) Graphical method for determining the roots of the equation $\tan\beta = \beta$.

as the conditions for minima. The first minimum occurs at $\theta = \pm\sin^{-1}\left(\frac{\lambda}{b}\right)$; the second minimum at $\theta = \pm\sin^{-1}\left(\frac{2\lambda}{b}\right)$, etc. Since $\sin\theta$ cannot exceed unity, the maximum value of m is the integer which is less than (and closest to) $\frac{b}{\lambda}$.

The positions of minima can directly be obtained by simple qualitative arguments. Let us consider the case $m = 1$. The angle θ satisfies the equation

$$b\sin\theta = \lambda \tag{18.13}$$

We divide the slit into two halves as shown in Fig. 18.5. Consider two points A and A' separated by a distance $b/2$. Clearly the path difference between the disturbances (reaching the point P) emanating from A and A' is $\frac{b}{2}\sin\theta$ which in this case is $\frac{\lambda}{2}$. The corresponding phase difference will be π and the resultant disturbance will be zero. Similarly, the disturbance from the point B will be cancelled by the disturbance reaching from the point B'. Thus, the resultant

* Equation (18.6) represents the amplitude distribution due to the interference of n point sources. Thus, for $n = 2$, the amplitude E_θ becomes $\cos\phi/2$ giving rise to $\cos^2\phi/2$ intensity distribution [cf. Eq. (14.13) of Chapter 14]. Notice that if we have a large number of equidistant sources oscillating in phase, then the propagation is only in cetain directions where the displacements add up in phase.

** We may mention here that in the limit $n \to \infty$ and $a \to 0$ the product na tends to a finite limit.

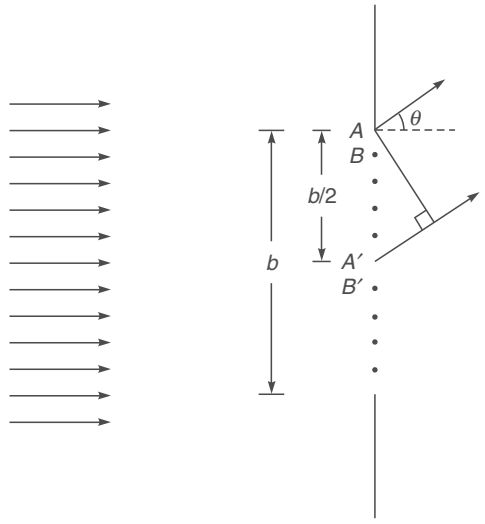


Fig. 18.5 The slit is divided into two halves for deriving the condition for the first minimum.

disturbance due to the upper half of the slit will be canceled by the disturbances reaching from the lower half and the resultant intensity will be zero. In a similar manner when

$$b \sin \theta = 2\lambda \qquad \textbf{(18.14)}$$

we divide the slit into four parts; the first and second quarters cancelling each other and the third and fourth quarters cancelling each other. Similarly when $m = 3$, the slit is divided into six parts and so on.

In order to determine the positions of maxima, we differentiate Eq. (18.10) with respect to β and set it equal to zero. Thus

$$\frac{dI}{d\beta} = I_0 \left[\frac{2 \sin \beta \cos \beta}{\beta^2} - \frac{2 \sin^2 \beta}{\beta^3} \right] = 0$$

or

$$\sin \beta [\beta - \tan \beta] = 0 \qquad \textbf{(18.15)}$$

The condition $\sin \beta = 0$, or $\beta = m\pi$ ($m \neq 0$) correspond to minima. The conditions for maxima are roots of the following transcendental equation

$$\tan \beta = \beta \quad \text{(maxima)} \qquad \textbf{(18.16)}$$

The root $\beta = 0$ corresponds to the central maximum. The other roots can be found by determining the points of intersections of the curves $y = \beta$ and $y = \tan \beta$ [see Fig. 18.4(b)]. The intersections occur at $\beta = 1.43\pi$, $\beta = 2.46\pi$, etc., and are known as the first maximum, the second maximum, etc. Since

$$\left[\frac{\sin (1.43\pi)}{1.43\pi} \right]^2$$

is about 0.0496, the intensity of the first maximum is about 4.96% of the central maximum. Similarly, the intensities of the second and third maxima are about 1.68% and 0.83% of the central maximum, respectively.

Example 18.1 A parallel beam of light is incident normally on a narrow slit of width 0.2 mm. The Fraunhofer diffraction pattern is observed on a screen which is placed at the focal plane of a convex lens whose focal length is 20 cm. Calculate the distance between the first two minima and the first two maxima on the screen. Assume $\lambda = 5 \times 10^{-5}$ cm and that the lens is placed very close to the slit.

Solution:

$$\frac{\lambda}{b} = \frac{5 \times 10^{-5}}{2 \times 10^{-2}} = 2.5 \times 10^{-3}$$

Now, the conditions for diffraction minima are given by $\sin \theta = m\lambda/b$. We assume θ to be small (measured in radians) so that we may write $\sin \theta \approx \theta$ (an assumption which will be justified by subsequent calculations); thus, on substituting the value of λ/b, we get

$$\theta \simeq 2.5 \times 10^{-3} \text{ and } 5 \times 10^{-3} \text{ radians}$$

as the angles of diffraction corresponding to the first and second minima, respectively. Notice that since

$$\sin (2.5 \times 10^{-3}) = 2.4999973 \times 10^{-3}$$

the error in the approximations $\sin \theta \simeq \theta$ is about 1 part in a million! These minima will be separated by a distance $(5 \times 10^{-3} - 2.5 \times 10^{-3}) \times 20 = 0.05$ cm on the focal plane of the lens. Similarly, the first and second maxima occur at

$$\beta = 1.43\pi \quad \text{and} \quad 2.46\pi$$

respectively. Thus,

$$b \sin \theta = 1.43\lambda \quad \text{and} \quad 2.46\lambda$$

or

$$\sin \theta = 1.43 \times 2.5 \times 10^{-3} \quad \text{and} \quad 2.46 \times 2.5 \times 10^{-3}$$

Consequently, the maxima will be separated by the distance given by

$$(2.46 - 1.43) \times 2.5 \times 10^{-3} \times 20 \simeq 0.05 \text{ cm}$$

Example 18.2 Consider, once again, a parallel beam of light ($\lambda = 5 \times 10^{-5}$ cm) to be incident normally on a long narrow slit of width 0.2 mm. A screen is placed at a distance of 3 m from the slit. Assuming that the screen is so far away that the diffraction is essentially of the Fraunhofer type, calculate total width of the central maximum.

Solution: As in Example 18.1, the first minimum occurs at $\theta \simeq 2.5 \times 10^{-3}$ radians; thus the total width of the central maximum is approximately given by